

Question number	Answer	Mark
1(a)(i)	A -70 mV B is not correct as -78 mV is the potential when the axon is hyperpolarised C is not correct as 38 mV is the action potential D is not correct as 108 mV is the change in potential from resting to action potential	(1)

Question number	Answer	Mark
1(a)(ii)	C voltage-gated potassium ion channel only A is not correct as the voltage-gated sodium ion channels will be closed and the potassium ion channels open at Y B is not correct as the voltage-gated sodium ion channels will be closed and the potassium ion channels open at Y D is not correct as the voltage-gated sodium ion channels will be closed and the potassium ion channels open at Y	(1)

Question number	Answer	Mark
1(a)(iii)	B hyperpolarised A is not correct as the membrane at Z is hyperpolarised C is not correct as the membrane at Z is hyperpolarised D is not correct as the membrane at Z is hyperpolarised	(1)

Question number	Answer	Mark
1(a)(iv)	B one A is not correct as only the third statement is correct. C is not correct as only the third statement is correct D is not correct as only the third statement is correct	(1)

Question number	Answer	Additional guidance	Mark
1(b)	An explanation that includes the following points: • because ATP is used by the sodium-potassium pump (1) • to move sodium ions and potassium ions across the cell membrane (1) • to establish concentration gradients for sodium ions and potassium ions (1)		